

What Actually Moves the Needle on BreakHis? A Patient-Level Audit of Magnification, Contrastive Learning, Retrieval and Pretraining

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Abstract. BreakHis dataset is widely used to benchmark breast histopathology classification models, but patient-level data leakage and unconstrained thresholds often make reported gains hard to interpret. Using a common Swin+FPN encoder under a leakage-free, patient-clustered, subtype-stratified protocol, we evaluate four proposed enhancements: magnification conditioning, supervised contrastive learning, retrieval-augmented memory, and pathology-specific pretraining. Across 16 held-out patients, image-level AUC stays within a one-point band for all of the methods, with every paired comparison showing negligible gains. Yet each method affects the model differently: magnification conditioning acts as a logit offset; contrastive learning improves representation geometry but not accuracy until that offset is removed; pathology-pretraining shows the same dissociation; and retrieval requires error-diversifying evidence, but even decorrelated neighbours must outperform the classifier already operating on the same feature space. These results show that on small histopathology benchmarks, architectural changes can alter representations and decision behaviour without improving patient-level discrimination. Evaluation protocol therefore plays a central role in determining what reported performance actually measures.

Keywords: Histopathology · Data Leakage · Patient Level Split · Swin Transformer · Feature Pyramid Network · Supervised Contrastive Learning · Retrieval-Augmented Classification

1 Introduction

Breast cancer is the most commonly diagnosed cancer in women worldwide, and the second most common cancer overall: 2.3 million new cases and 666,103 deaths in 2022, a quarter of all female cancer diagnoses [2]. India shows the gap between incidence and outcome starkly - third in new cases (192,020) but first in deaths (98,337) worldwide - a mismatch driven by screening and treatment access, not disease biology. The global count is projected to pass 6 million by 2050. The bottleneck is not diagnosis in principle; it is pathologist throughput.

Deep learning addresses that bottleneck directly. It learns diagnostic features from tissue images without hand-crafted descriptors, cuts the inter-observer variability of manual review, and scales to caseloads that outstrip pathologist availability, a real constraint in a country like India. Convolutional networks and vision transformers now report 94–99% accuracy on benchmarks such as BreaKHis [31]. What remains unresolved is not raw accuracy but generalization across scanners and stains, interpretability, and evaluation itself.

That last point is not a theoretical concern for BreaKHis [24], the standard benchmark for classification tasks. Reported accuracies on it are rarely comparable across papers, because images from the same patient routinely leak across the train/test split and inflate the score [3]. Leakage-checked re-releases now exist for exactly this reason [11], and a recent patient-isolated benchmark traces much of the field’s reported architectural gap to the partition, not the architecture [23]. A benchmark this compromised cannot carry the comparisons made on it.

The classification backbone used here is background, not a contribution. CNNs have a limited receptive field; plain vision transformers trade that limit for quadratic-cost attention over a single scale. The Swin Transformer fixes both by computing attention inside shifted local windows, producing CNN-like multi-scale maps at linear cost [17]. A Feature Pyramid Network then fuses those stages top-down, so fine local detail and broad tissue context reach the classifier together [16] - useful here because nuclei, glands and tumour regions in a slide sit at very different scales.

Beyond classification, pathology has a separate line of work on retrieval. Whole-slide search systems index tissue and return similar cases [12,4,28], and a newer idea, retrieval-augmented classification, feeds nearest neighbours back into the decision itself [18] - an idea pathology has only begun to adopt [21,30]. Whether such a memory adds anything beyond what a well-trained encoder already knows is still open [15].

2 Related Work

BreaKHis and evaluation integrity. BreaKHis [24] is the standard benchmark for this task. Early work classified it directly with convolutional networks [25]; surveys track the literature that followed [31], and recent papers add contrastive objectives on top of it [20]. Reported numbers on the dataset are hard to compare, though. Bussola et al. measured accuracy inflation up to 41% from image-level partitioning of histopathology slides [3]. Leakage-checked re-releases of BreaKHis exist for that reason [11], and a patient-isolated benchmark traces much of the field’s reported architectural gap to the partition rather than the architecture [23]. The problem is not specific to this dataset: outperformance claims in medical imaging AI often fail a stricter multi-centre validity check [8], and leakage of

this kind is a documented driver of the wider reproducibility crisis in machine-learning-based science [13].

Magnification and contrastive learning. BreakHis images are captured at four magnifications, so classification methods split into magnification-specific and magnification-unified approaches [26]. Bayramoglu et al. trained a multi-task CNN to predict malignancy and magnification jointly [1], Gupta and Bhavsar measured how much the choice of magnification affects accuracy [10], and recent work targets magnification-invariant representations through domain generalization [9]. Contrastive learning is a separate, widely used tool in this space. Self-supervised contrastive pretraining [7] and its supervised variant [14] both underpin large pathology encoders such as CTransPath [29], as well as task-specific models trained directly on BreakHis [20].

Retrieval and foundation models. Histopathology has its own line of retrieval work. Yottixel indexes whole-slide images as compact barcodes [12], SISH builds a self-supervised, scalable index [4], and RetCCL learns features for slide-level search [28]; an independent validation found the performance of such systems highly sensitive to evaluation design [15]. Retrieval-augmented classification, which feeds nearest neighbours back into the decision itself, is established in general vision [18] and has entered pathology mainly through retrieval-augmented generation [21,30]. Large pretrained encoders now dominate reported results in the field: UNI [6], Virchow [27], CTransPath [29], the hierarchical self-supervised HIPT [5], and the vision-language model CONCH [19]. How much advantage they give as feature extractors varies by task and evaluation protocol [22].

3 Methodology

3.1 Dataset

We use the BreakHis dataset [24]: 7,909 H&E-stained histopathology images from 82 patients, each a three-channel RGB PNG of 700×460 pixels. The binary label is derived from eight finer subtypes: four benign (adenosis, fibroadenoma, phylloides tumor, tubular adenoma) and four malignant (ductal, lobular, mucinous, papillary carcinoma). Table 1 gives the full composition of the dataset alongside the splits as mentioned in Section 3.2. The subtype distribution is long-tailed at both the patient and the image level.

Each sample is imaged at four magnifications ($40\times$, $100\times$, $200\times$, $400\times$; 1,995 / 2,081 / 2,013 / 1,820 images). Low magnification shows tissue architecture; high magnification shows nuclear detail. Every patient contributes images at all four scales, so the same tissue region repeats across magnifications, producing highly correlated images within a patient. This redundancy, plus large inter- and intra-class variability and the long-tailed subtype distribution, makes patient-level partitioning a key methodological concern, not just an implementation detail.

Table 1: Composition of BreaKHis and the two partitions produced by the split policy.

Tumour subtype (code)	Full dataset			Held-out test		CV pool	
	Pat.	n	%	Pat.	Images	Pat.	Images
<i>Benign</i>							
Adenosis (A)	4	444	5.6	1	132	3	312
Fibroadenoma (F)	10	1,014	12.8	2	194	8	820
Phyllodes tumour (PT)	3	453	5.7	0	0	3	453
Tubular adenoma (TA)	7	569	7.2	1	132	6	437
<i>Benign total</i>	<i>24</i>	<i>2,480</i>	<i>31.4</i>	<i>4</i>	<i>458</i>	<i>20</i>	<i>2,022</i>
<i>Malignant</i>							
Ductal carcinoma (DC)	38	3,451	43.6	8	740	30	2,711
Lobular carcinoma (LC)	5	626	7.9	1	125	4	501
Mucinous carcinoma (MC)	9	792	10.0	2	240	7	552
Papillary carcinoma (PC)	6	560	7.1	1	90	5	470
<i>Malignant total</i>	<i>58</i>	<i>5,429</i>	<i>68.6</i>	<i>12</i>	<i>1,195</i>	<i>46</i>	<i>4,234</i>
All	82	7,909	100.0	16	1,653	66	6,256

3.2 Splitting Protocol

Patient-level partitioning. Each patient contributes 60–235 images of one specimen, so we treat the patient, not the image, as the atomic partitioning unit: image-level splits leak near-duplicate views across train/test and inflate accuracy by up to 41% [3] (Section 2). Of the 82 patients, 16 (19.5%) are held out permanently, untouched by training, validation, model selection, or calibration; the remaining 66 form five class-stratified, patient-disjoint folds of 52–53 training and 13–14 validation patients. Splits are generated once under seed 42, serialized with their policy to a JSON artifact, and reloaded verbatim; disjointness is checked at generation, at every load, and again when the retrieval bank is built.

Subtype-stratified test policy. An earlier version stratified the held-out set by binary class only. The subtype distribution is long-tailed, so this let rare subtypes clump: one draw took three fibroadenoma patients and the largest slide, a 235-image phyllodes case, but no adenosis and no papillary carcinoma, and image-level test accuracy sat near 82% for every model we trained. We replaced it with a rule that stratifies the held-out set over all eight subtypes: each gets a quota by largest-remainder allocation and a seeded within-subtype draw (counts in Table 1). Two guards keep the estimate representative. *Subtype reservation* keeps phyllodes tumor, only three patients in BreaKHis, out of the test set, since holding one out would starve training of a rare morphology and let a single high-variance patient swing the held-out score. An *image-share cap* of $f = 0.12$ of the expected test budget ($7,909 \times 16/82 \approx 1,543$ images) routes the 201-image lobular patient SOB_M_LC-14-15570 to the pool, so no single slide dominates the image-level metric. Folds stay class-stratified only, since the scarcest subtypes cannot be stratified across five.

A rule, not a search. A hand-tuned test split invites the objection that it was searched for a favorable result. Four facts argue against that: the rule is applied once under a fixed seed; its quota and redirected-patient list are serialized into the splits file, not described after the fact; the split was frozen before any model here was trained; and the superseded class-only split is retained rather than deleted, so the change is auditable. Crucially, the new split removes the artifact that pinned accuracy at a near-constant value regardless of model, making the held out test-metric harder to saturate not easier. The magnification-unified setting is part of the protocol, not a post hoc choice. Finally, a single held-out patient is worth 6.25 percentage points of patient-level accuracy: the effective sample size here is the patient count, not the image count, so we report both image- and patient-level metrics and pair every comparison with patient-clustered bootstrap intervals.

3.3 Pre-Processing

Images are resized to 224×224 and normalized with ImageNet statistics to match the pretrained Swin backbone. Training augmentation is deliberately light: horizontal/vertical flips ($p = 0.5$), rotation ($\pm 15^\circ$), and colour jitter (brightness, contrast, saturation 0.1, hue 0.02); hue is nearly frozen and random erasing is off because H&E stain colour is diagnostic signal, not noise. Validation, testing, and memory-bank encoding use deterministic resizing and normalization only, so stored descriptors match inference exactly. Class imbalance is handled by inverse-frequency weighted cross-entropy rather than oversampling, leaving Table 1’s image counts untouched.

3.4 Baseline Architecture

Backbone and pyramid. An ImageNet-pretrained Swin Transformer-Tiny [17] maps $I \in \mathbb{R}^{3 \times 224 \times 224}$ through four stages $\{F_1, \dots, F_4\}$ of 96, 192, 384, 768 channels at strides 4, 8, 16, 32, each stage alternating window and shifted-window blocks:

$$\begin{aligned} \hat{z}^\ell &= \text{W-MSA}(\text{LN}(z^{\ell-1})) + z^{\ell-1}, & z^\ell &= \text{MLP}(\text{LN}(\hat{z}^\ell)) + \hat{z}^\ell, \\ \hat{z}^{\ell+1} &= \text{SW-MSA}(\text{LN}(z^\ell)) + z^\ell, & z^{\ell+1} &= \text{MLP}(\text{LN}(\hat{z}^{\ell+1})) + \hat{z}^{\ell+1}, \end{aligned}$$

with window attention $\text{Attn}(Q, K, V) = \text{softmax}\left(QK^\top/\sqrt{d} + B\right)V$. A Feature Pyramid Network [16] then unifies the four stages to $P_1, \dots, P_4$ of 256 channels each through 1×1 lateral convolutions L_i , 3×3 output convolutions c_i , and a top-down pathway:

$$P_4 = c_4(L_4(F_4)), \quad P_i = c_i(L_i(F_i) + \text{Up}(P_{i+1})), \quad i = 3, 2, 1.$$

Fused descriptor. Every scale, not only the deepest, should reach the heads:

$$z = \text{Concat}(\text{GAP}(P_1), \dots, \text{GAP}(P_4)) \in \mathbb{R}^{1024}.$$

Holding z fixed across configurations makes each enhancement of Section 3.5 a controlled modification of one model rather than a separate model. It is also

defined *before* any magnification information enters the network, which is what makes it usable as a retrieval key in Section 3.5.

Classification head. Dropout ($p = 0.1$) followed by a single linear layer $\mathbb{R}^{1024} \rightarrow \mathbb{R}^2$. The head is deliberately shallow, so measured differences reflect the representation rather than the capacity of the classifier.

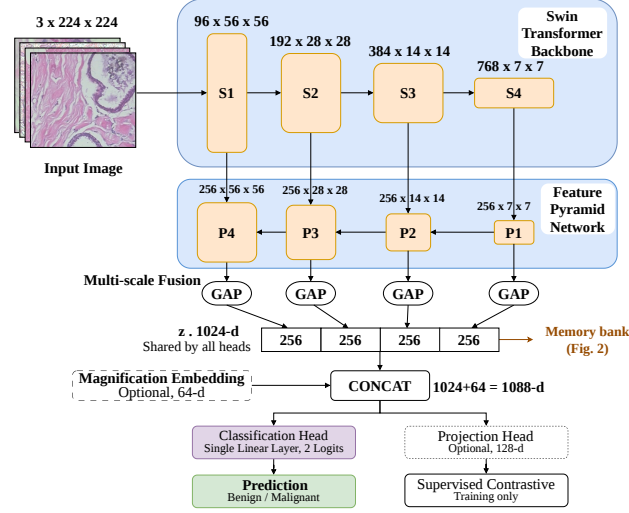


Fig. 1: Baseline architecture: a Swin-Tiny backbone and a feature pyramid produce the 1024-dimensional descriptor z shared by every configuration, with the two switched components of Table 2 drawn dashed and shapes as channels $\times$ spatial.

3.5 Candidate Enhancements

Three enhancements are commonly proposed for histopathology classification: explicit magnification conditioning, supervised contrastive learning, and non-parametric retrieval from a memory of previously seen training cases. We treat each as a hypothesis to test, not a component assumed to help. The first two are trained variants of one model and form the 2×2 factorial of Table 2, which isolates supervised contrastive learning applied *without* magnification conditioning (Exp3n). The third is a post-hoc module on a frozen encoder and is described separately.

Table 2: The four trained configurations form a 2×2 factorial over the two enhancements. All four share the backbone, the fused descriptor z , the classification head and the training schedule, and differ only in the two switched components.

Configuration	Magnification embedding	SupCon	Training objective
Exp1	—	—	$\mathcal{L}_{CE}$
Exp2	✓	—	$\mathcal{L}_{CE}$
Exp3	✓	✓	$\mathcal{L}_{CE} + \lambda \mathcal{L}_{SupCon}$
Exp3n	—	✓	$\mathcal{L}_{CE} + \lambda \mathcal{L}_{SupCon}$

Magnification Embedding: A model trained jointly over all four magnifications can be told which one it is looking at. A learnable table $E \in \mathbb{R}^{4 \times 64}$, indexed by the magnification $m \in \{40\times, 100\times, 200\times, 400\times\}$, is concatenated to the fused descriptor,

$$z' = [z; E_m] \in \mathbb{R}^{1088},$$

and z' replaces z at the heads; E is learned with the rest of the network. Because the term depends only on m and enters after all spatial processing, it can move the logits only by a per-magnification offset.

Supervised Contrastive Learning: Cross-entropy constrains only the decision boundary, so we add the supervised contrastive term of Khosla et al. [14], which pulls same-class samples of a mini-batch together and pushes the other class apart. A two-layer projection head maps the descriptor onto a 128-dimensional unit sphere,

$$p = \text{Norm}(W_2 \sigma(W_1 z')),$$

with $W_1 : 1024 \rightarrow 512$, or $1088 \rightarrow 512$ when magnification conditioning is on, $W_2 : 512 \rightarrow 128$, σ the ReLU activation and Norm the L2 normalization. $\mathcal{L}_{\text{SupCon}}$ acts on p alone, classification acts on z' , and the projection head is discarded after training.

Retrieval-Augmented Memory: The two enhancements above change what the encoder learns. A third leaves it untouched and supplies evidence at inference time, so a prediction can be backed by similar cases already seen. We test it in its most controlled form (Fig. 2). The module is *post-hoc* and two-stage: the encoder is trained, frozen, and then used both to write the memory and to answer queries, and the only trainable component is a 147-parameter fusion gate. No retrieval term enters the training objective and nothing in Fig. 1 changes, so any measured difference belongs to the module alone. Exp3n supplies the frozen encoder: its descriptor is the one defined without magnification conditioning.

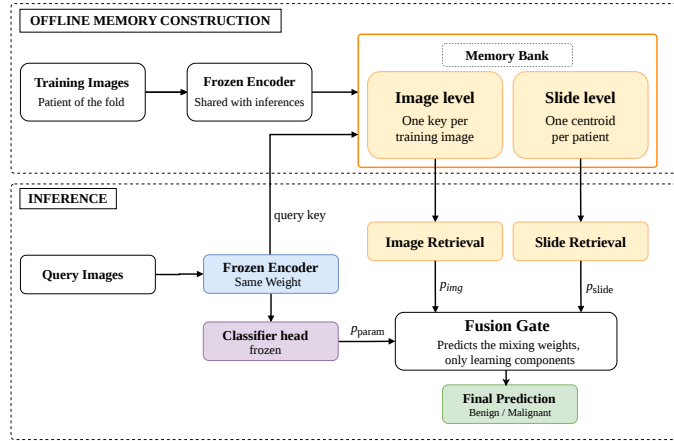


Fig. 2: Retrieval-augmented memory: the frozen encoder writes an image level and a slide level per fold, the two are ranked *independently* by cosine similarity, and a 147-parameter gate blends them with the classifier’s posterior.

Memory. One memory per fold, built from that fold’s training patients only. Every training image contributes a key $k_i = z_i / \|z_i\|_2 \in \mathbb{R}^{1024}$ with its label, patient identity, tumour subtype and magnification, where z_i is the pre-magnification descriptor of Section 3.4. The same store holds one slide-level row per training patient, the renormalized mean of that patient’s keys, so a query can be matched against single fields and whole specimens; fold 0 gives 4,541 image rows and 52 slide rows. Patient disjointness is asserted at write, at load, and again for every query.

Retrieval. A query key q is ranked against the two levels *independently* by cosine similarity $s(q, k) = q^\top k$ on unit-norm vectors; a single ranking over their union is dominated by the far more numerous image rows, so the levels share one store but never a ranking. Image-level search is restricted to the query’s own magnification and returns $K = 15$ neighbours under a cap of 3 per bank patient, so one large slide cannot carry a vote alone; slide-level search ignores magnification and returns $K = 5$, one per patient by construction. A query never retrieves its own patient. Each neighbourhood $\mathcal{N}$ votes by a temperature-scaled softmax over its similarities,

$$p_{\text{level}}(y \mid q) = \sum_{i \in \mathcal{N}} \frac{\exp(s(q, k_i)/\tau)}{\sum_{j \in \mathcal{N}} \exp(s(q, k_j)/\tau)} \mathbb{I}[y_i = y], \quad \tau = 0.07.$$

Retrieved subtypes are recorded for interpretability and never used as vote weights.

Fusion. The two votes are combined with the classifier’s own posterior as a convex combination

$$p_{\text{final}} = w_{\text{param}} p_{\text{param}} + w_{\text{img}} p_{\text{img}} + w_{\text{slide}} p_{\text{slide}},$$

$$(w_{\text{param}}, w_{\text{img}}, w_{\text{slide}}) = \text{softmax}(g(\phi)),$$

where $g : \mathbb{R}^5 \rightarrow \mathbb{R}^{16} \rightarrow \mathbb{R}^3$ is the gate and $\phi = [p_{\text{param}}, |p_{\text{param}} - 0.5|, 2|p_{\text{img}} - 0.5|, \max_i s(q, k_i), \text{distinct-patient fraction of } \mathcal{N}]$ holds confidence and neighbourhood statistics rather than image features, so the blend can vary with how informative the neighbourhood is. The gate is fitted *once* on pooled out-of-fold validation predictions and shared by all folds, since fitting it per fold overfits the validation signal. Predicted weights also let the module fall back entirely on p_{param} where retrieval adds nothing, so the comparison tests the enhancement and not the blend.

3.6 Training Protocol

Optimization: All models use AdamW ($\beta_1 = 0.9$, $\beta_2 = 0.999$, initial learning rate 3×10^{-4} , weight decay 0.05) with cosine annealing to a minimum of 1×10^{-6} . Mini-batches of 16 with gradient accumulation over two steps give an effective batch of 32, gradients are clipped to norm 1.0, and mixed precision (AMP) is used throughout. Exp1 runs for 10 epochs and every later configuration for 15, a budget fixed for all runs rather than tuned, since validation performance plateaus well inside it. Checkpointing and early stopping both track the best image-level validation accuracy, with a patience of 10 epochs. Every run uses the same budget on one NVIDIA GeForce RTX 3050 Laptop GPU with 4 GB of VRAM, the constraint behind mixed precision and gradient accumulation.

Loss Functions: Every configuration is trained with class-weighted cross-entropy over the posterior $\hat{p}$, the weights being inverse to the image counts n_c :

$$\mathcal{L}_{\text{CE}} = - \sum_{c \in \{0,1\}} w_c y_c \log \hat{p}_c, \quad w_c \propto 1/n_c.$$

where y_c is the one-hot ground-truth label, $\hat{p}_c$ the model’s predicted probability for class c , and n_c the number of training images of class c .

Configurations with supervised contrastive learning add a term over the L2-normalized projections p of a mini-batch I , where $P(i)$ is the set of samples sharing the label of i :

$$\mathcal{L}_{\text{SupCon}} = \sum_{i \in I} \frac{-1}{|P(i)|} \sum_{q \in P(i)} \log \frac{\exp(p_i^\top p_q / \tau)}{\sum_{a \in I \setminus \{i\}} \exp(p_i^\top p_a / \tau)}.$$

where q indexes the other same-label samples and a the rest of the batch, and τ is the softmax temperature (value below).

The trained objective is their weighted sum (Table 2):

$$\mathcal{L} = \mathcal{L}_{\text{CE}} + \lambda \mathcal{L}_{\text{SupCon}}, \quad \lambda = 0.4, \quad \tau = 0.07.$$

Classification always reads z , never p . These are the only objectives: the retrieval module is fitted after the encoder is frozen and adds no training loss.

Threshold Calibration: Accuracy under class imbalance is sensitive to the decision threshold, so comparing models at a default of 0.5 confounds discrimination with calibration. Each threshold is instead chosen on pooled out-of-fold validation predictions and locked before the test set is touched, so every model is read at its own operating point and no test information enters the choice.

3.7 Evaluation and Statistical Analysis

Classification. Every configuration, the frozen CTransPath probe included, is scored on the held-out set of Section 3.2 with one metric set: accuracy, area under the receiver operating characteristic curve (AUC), sensitivity and specificity, each at the image level and at the patient level, the latter from a patient’s mean image probability. AUC is threshold-free; the rest are read at 0.5 and at the locked out-of-fold threshold of Section 3.6, with a test-fitted threshold used only as an oracle bound. Point estimates are five-fold means and the fold ensemble, which averages the five probability vectors before thresholding. Paired differences in AUC and accuracy carry 95% intervals from a patient-clustered bootstrap over the 16 held-out patients, 2,000 resamples, dropping any resample without a benign patient. Geometry is read from the cosine silhouette of z under the binary and the eight-subtype labelling, the accuracy of a patient-blocked leave-one-out 15-neighbour vote, the share of those neighbours at the query’s magnification (chance 0.25), and the same vote’s accuracy at predicting magnification instead of class.

Retrieval. The memory is read on four axes under that same metric set. *Streams:* p_{param} , p_{img} , p_{slide} and the fusion p_{final} scored separately, plus how many decisions

the fusion changes and in which direction. *Complementarity*: the correlation between p_{img} and p_{param} , and the AUC of p_{img} on the images the classifier gets wrong. *Gate*: mean predicted weights, over correct and incorrect images separately, refitted leave-one-fold-out when configurations are compared. *Neighbourhood*: mean top-1 cosine similarity, neighbour overlap between two memories, distinct bank patients per query, the same-magnification rate of the active route, and subtype lift, the chance a neighbour shares the query’s subtype minus that subtype’s rate in the bank. Key and configuration ablations use pooled out-of-fold validation only, reporting each key space’s effective rank beside its retrieval AUC.

3.8 Foundation Model Baseline

A fourth candidate is pathology-specific pretraining itself, included as an external control rather than a rung of the ladder. CTransPath [29], a Swin-Tiny encoder contrastively pretrained on about 15 million histopathology patches, matches our backbone in architecture family and capacity, so the comparison isolates the pretraining corpus from model scale. We keep it frozen, extract features once for all 7,909 images under the same deterministic transform and patient-level splits, and fit one linear head per fold, with class weighting, optimizer, threshold calibration and metrics held identical. Any difference that remains is down to what the encoder was pretrained on.

4 Results & Discussion

Each verdict below is paired with a mechanism, not just a delta. Sections 4.1–4.5 read the held-out test set (16 patients, 1,653 images); Section 4.6 is a pooled out-of-fold screen (66 patients, 6,256 images).

Table 3: Held-out benchmark (16 patients, 1,653 images): five-fold mean point estimates, patient-clustered bootstrap differences on the fold ensemble, and thresholds locked out of fold before the test set was touched.

Configuration	Train. params	Acc. @0.5	Locked thr.	Acc. @ thr.	Image AUC	Sens.	Spec.	Pat. acc.
Exp1 baseline	30.8 M	0.9054	0.450	0.9064	0.9559	0.9242	0.8563	0.9875
Exp2 + mag	30.8 M	0.8974	0.428	0.9016	0.9550	0.9054	0.8764	0.9625
Exp3 + mag + SupCon	30.8 M	0.8858	0.360	0.8955	0.9607	0.8894	0.8764	0.9375
Exp3n + SupCon	30.8 M	0.8986	0.540	0.8981	0.9593	0.9193	0.8445	0.9750
+ retrieval memory	+147	0.9057	0.692	0.8874	0.9561	—	—	0.9750
CTransPath (frozen)	1,538	0.8580	0.209	0.8985	0.9668	0.8190	0.9590	0.9375

Paired difference	Δ image AUC [95% CI]	Δ image acc. [95% CI]
Exp2 – Exp1	+0.0032 [–0.0046, +0.0141]	+0.0005 [–0.0064, +0.0057]
Exp3 – Exp1	+0.0038 [–0.0014, +0.0117]	–0.0023 [–0.0154, +0.0092]
Exp3n – Exp1	+0.0029 [–0.0041, +0.0112]	+0.0083 [–0.0135, +0.0344]
Exp3n – Exp3	–0.0009 [–0.0130, +0.0090]	+0.0106 [–0.0176, +0.0475]
+ retrieval – Exp3n	–0.0003 [–0.0009, +0.0007]	–0.0254 [–0.0599, +0.0028]
CTransPath – Exp1	+0.0028 [–0.0490, +0.0573]	–0.0063 [–0.0836, +0.0755]

4.1 Overall Comparison

Table 3 asks whether the accuracy column can settle the comparison. It cannot. Held-out AUC sits in a narrow band, while accuracy at a fixed 0.5 cut spreads four times wider. Locking each model to its own out-of-fold threshold shrinks that spread to about one percentage point, some 18 images. None of the six paired contrasts, in AUC or in accuracy, is distinguishable from zero, and patient accuracy saturates at 16/16. Discrimination is flat across the ladder; what moves is where the boundary sits. A comparison at 0.5 therefore measures calibration, not capability.

4.2 Magnification Conditioning

Fig. 3 confirms what Section 3.5 predicts: the block moves the logits by one scalar per magnification. The four offsets are monotone in zoom but span only a seventh of the ± 4.22 image-driven spread. Zeroing the block, replacing it by its mean or forcing every image to one magnification all return the same AUC, 0.96050 against the deployed 0.96068. The limit is the read-out, not redundancy: the descriptor recovers magnification only modestly above chance. Even four per-magnification thresholds fitted on the test set buy at most one point, which bounds the idea, not just our implementation. The block also locks the representation to magnification (Table 4), so it cannot serve as a retrieval key: a bias term dressed up as a change to the representation.



Fig. 3: Magnification audit (Exp3, five-fold means, test set): the four learned per-zoom logit offsets against the shaded image-driven spread (a), and image AUC under five head inputs (b).

4.3 Supervised Contrastive Learning

Table 4 separates the contrastive term from the magnification block. The term raises binary silhouette whether or not the block is present, and the block lowers it either way. Removing the block recovers image and patient accuracy, restores sensitivity, leaves AUC unchanged and un-locks the space, and Exp3n is the only configuration whose locked threshold sits near 0.5. What the objective does not fix is subtype structure. Silhouette is negative in every space and every fold,

with papillary, mucinous and lobular carcinoma smeared through the ductal mass (Fig. 4, in all twenty cells). A representation that separates classes but not subtypes is exactly where a non-parametric memory should help.

Table 4: Representation geometry and operating point over the 2×2 (five-fold means, test set); the same-magnification rate is read against a chance rate of 0.25.

Configuration	Mag.	SupCon	Binary silhouette	Same-mag. rate	Locked thr.	Img. acc. @ thr.
Exp1	—	—	0.623	0.314	0.450	0.9064
Exp2	✓	—	0.538	—	0.428	0.9016
Exp3	✓	✓	0.569	1.000	0.360	0.8955
Exp3n	—	✓	0.680	0.342	0.540	0.8981

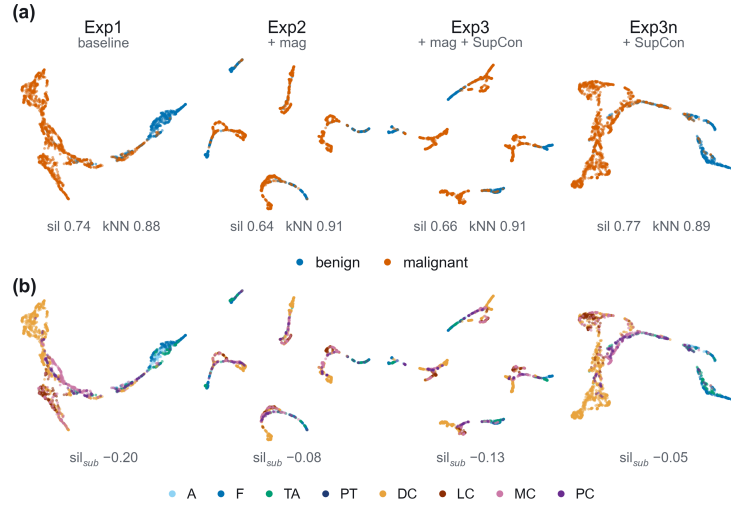


Fig. 4: UMAP of the test set under the fold-0 model, coloured by class (a) and by the eight subtypes (b); the per-panel values are single-fold, and Table 4’s five-fold means carry every claim.

4.4 Retrieval Memory with a Shared Encoder

The held-out result is a null (Table 3): AUC unchanged, accuracy at the pre-registered threshold worse in all five folds, about one per cent of predictions changed in both directions. The reading is threshold-sensitive: at a fixed 0.5 cut the fusion beats the classifier in five folds out of five while losing AUC in all five. Only the pre-registered threshold is admissible.

Neither the gate nor the key is at fault. The gate shifts weight onto the memory exactly where the classifier errs, in every fold. Of forty-three key definitions spanning a 338-fold range of effective rank, none of the thirty-nine written by

the classifier’s own encoder out-discriminates it; only the slide-centroid level, at fifty centroids per fold, is significantly worse. The redundancy is semantic, not geometric, and it follows from the shared encoder. What the module delivers is evidence, not accuracy: retrieved neighbours match the query’s subtype well above base rate on patients it has never seen. That licenses a prediction. A bank written by an encoder that errs elsewhere should decorrelate far more.

4.5 Pathology-Specific Pretraining

A frozen pathology encoder with a linear head asks whether the ceiling is ours or the dataset’s. CTransPath is capacity-matched to our backbone and, with 1,538 trainable parameters, ties at its own locked threshold (Table 3). The shape of the tie matters more: the best AUC in the study, the worst accuracy at 0.5, the most extreme locked threshold measured. Section 4.1’s dissociation therefore reappears from a different architecture, corpus and objective. It is also wrong elsewhere: the two patients carrying most of the Swin ladder’s error are easy for it, and a mucinous carcinoma every Swin variant finds trivial is its worst failure. Two encoders, indistinguishable accuracy, wrong in different places: the untested condition of Section 4.4.

4.6 Retrieval Memory with an Un-shared Encoder

So we rebuilt the memory with the frozen foundation encoder, left the classifier untouched, and screened it out of fold only. The mechanism is confirmed as a three-point dose–response (Fig. 5a): error-decorrelation rises monotonically to an order of magnitude above the shared-encoder value, the neighbourhoods barely overlap, the gate triples the weight it keeps on the memory, and the memory’s own vote out-discriminates the classifier it was built to assist.

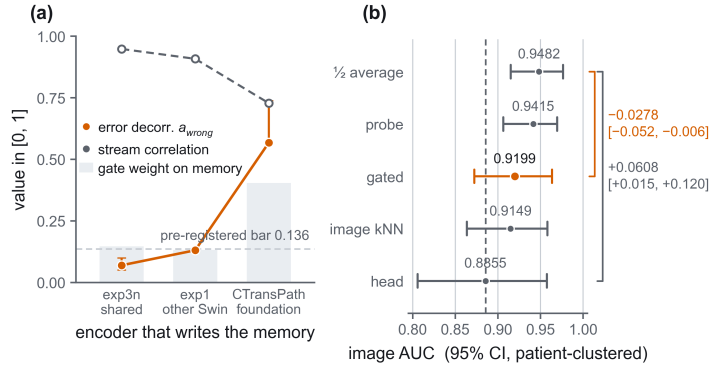


Fig. 5: The retrieval boundary condition, pooled out of fold: error-decorrelation, stream correlation and gate weight against the encoder that writes the memory (a), and image AUC of five probability streams with the two decisive paired contrasts (b).

It still loses, though not to the classifier: a plain probability average of the same two models beats the gated fusion by a margin whose interval excludes zero

(Fig. 5b, Table 5). Without that control the run would read as a +0.033 AUC gain. The gate is not the culprit: delete it, pick the blend by oracle, and the memory still falls short. The constraint is the read-out. On identical frozen features a cosine neighbourhood vote scores clearly below a 1,538-parameter linear layer, and the null survives seven routing variants, a hundred-fold temperature sweep and both threshold conventions. One by-product survives: subtype lift is the highest recorded here. Error-decorrelation is necessary, but not sufficient.

Table 5: Probability streams under a CTransPath-written memory, pooled out of fold, with patient-clustered differences against the plain two-model average p_{ens} .

Stream	Definition	Img. AUC	Δ vs p_{ens} [95% CI]
p_{param}	Exp3n classifier, 30.8 M	0.8855	-0.0608 [-0.1201, -0.0152]
p_{img}	k NN vote, CTransPath bank	0.9149	—
p_{final}	gated fusion, 147 params	0.9199	-0.0278 [-0.0524, -0.0065]
p_{probe}	CTransPath + linear head	0.9415	—
p_{ens}	half-and-half average	0.9482	—
oracle blend	best fixed blend, gate deleted	0.9383	-0.0099

5 Conclusion

We tested four proposed enhancements as controlled add-ons to one Swin + FPN encoder on BreakHis: magnification conditioning, supervised contrastive learning, a retrieval-augmented memory, and pathology-specific pretraining. We used a leakage-free, patient-clustered protocol throughout. None moves the held-out result: image AUC stays within a one-point band, and no paired difference clears zero. What moves is the operating point. A fixed 0.5 threshold measures calibration, not capability, and both our ladder and an external frozen encoder confirm this.

These null results are structural, not incidental. Magnification conditioning is only a four-value logit offset. Contrastive learning always improves geometry, but only becomes accuracy once that offset is removed. The memory fails for a reason that generalizes: error-decorrelation is *necessary* for a retrieved neighbourhood to help but *not sufficient*, since its vote must beat a fitted layer on the same features. The one exception in the whole study is plain averaging of two independently pretrained encoders’ probabilities, out of fold. We do not claim it, but it shows our ceiling is per-encoder, not per-task. The next increment belongs to the read-out, a learned combiner over complementary encoders, not a better key or a larger bank. On a dataset this size, protocol decides what a reported number means, not architecture.

References

1. Bayramoglu, N., Kannala, J., Heikkilä, J.: Deep learning for magnification independent breast cancer histopathology image classification. In: Proceedings of the

- International Conference on Pattern Recognition (ICPR). pp. 2440–2445 (2016). <https://doi.org/10.1109/ICPR.2016.7900002>
2. Bray, F., Laversanne, M., Sung, H., Ferlay, J., Siegel, R.L., Soerjomataram, I., Jemal, A.: Global cancer statistics 2022: Globocan estimates of incidence and mortality worldwide for 36 cancers in 185 countries. *CA: a cancer journal for clinicians* **74**(3), 229–263 (2024)
 3. Bussola, N., Maggio, V., Marcolini, A., Jurman, G., Furlanello, C.: AI slipping on tiles: data leakage in digital pathology. arXiv preprint arXiv:1909.06539 (2019)
 4. Chen, C., Lu, M.Y., Williamson, D.F.K., Chen, T.Y., Schaumberg, A.J., Mahmood, F.: Fast and scalable search of whole-slide images via self-supervised deep learning. *Nature Biomedical Engineering* **6**, 1420–1434 (2022). <https://doi.org/10.1038/s41551-022-00929-8>
 5. Chen, R.J., Chen, C., Li, Y., Chen, T.Y., Trister, A.D., Krishnan, R.G., Mahmood, F.: Scaling vision transformers to gigapixel images via hierarchical self-supervised learning. In: *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. pp. 16144–16155 (2022)
 6. Chen, R.J., Ding, T., Lu, M.Y., Williamson, D.F.K., Jaume, G., Song, A.H., Chen, B., Zhang, A., Shao, D., Shaban, M., Williams, M., Oldenburg, L., Weishaupt, L.L., Wang, J.J., Vaidya, A., Le, L.P., Gerber, G., Sahai, S., Williams, W., Mahmood, F.: Towards a general-purpose foundation model for computational pathology. *Nature Medicine* **30**(3), 850–862 (2024). <https://doi.org/10.1038/s41591-024-02857-3>
 7. Chen, T., Kornblith, S., Norouzi, M., Hinton, G.: A simple framework for contrastive learning of visual representations. In: *Proceedings of the International Conference on Machine Learning (ICML)* (2020)
 8. Christodoulou, E., Reinke, A., André, P., Godau, P., Kalinowski, P., Houhou, R., Erkan, S., Sudre, C.H., Burgos, N., Boutaj, S., Cheplygina, V., Antonelli, M., Rieke, N., Cardoso, M.J., Simpson, A., Kopp-Schneider, A., Varoquaux, G., Colliot, O., Maier-Hein, L.: False promises in medical imaging AI? assessing validity of outperformance claims. arXiv preprint arXiv:2505.04720 (2025), <https://arxiv.org/abs/2505.04720>
 9. Ezuma, I., Medaiyese, O.: Magnification-invariant image classification via domain generalization and stable sparse embedding signatures. arXiv preprint arXiv:2604.25817 (2026), <https://arxiv.org/abs/2604.25817>
 10. Gupta, V., Bhavsar, A.: Breast cancer histopathological image classification: Is magnification important? In: *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition Workshops (CVPRW)*. pp. 769–776 (2017). <https://doi.org/10.1109/CVPRW.2017.107>
 11. Kabir, S.: BreakHis-IB: Leakage-checked splits across magnifications for robust BreakHis evaluation. *IEEE DataPort* (2025). <https://doi.org/10.21227/z4mc-k240>
 12. Kalra, S., Tizhoosh, H.R., Choi, C., Shah, S., Diamandis, P., Campbell, C.J., Pantanowitz, L.: Yottixel – an image search engine for large archives of histopathology whole slide images. *Medical Image Analysis* **65**, 101757 (2020). <https://doi.org/10.1016/j.media.2020.101757>
 13. Kapoor, S., Narayanan, A.: Leakage and the reproducibility crisis in machine-learning-based science. *Patterns* **4**(9), 100804 (2023). <https://doi.org/10.1016/j.patter.2023.100804>
 14. Khosla, P., Teterwak, P., Wang, C., Sarna, A., Tian, Y., Isola, P., Maschinot, A., Liu, C., Krishnan, D.: Supervised contrastive learning. In: *Advances in Neural Information Processing Systems*. vol. 33, pp. 18661–18673 (2020)

15. Lahr, I., Alfasly, S., Nejat, P., Khan, J., Kottom, L., Kumbhar, V., Alsaafin, A., Shafique, A., Hemati, S., Alabtah, G., Comfere, N., Murphree, D., Mangold, A., Yasir, S., Meroueh, C., Boardman, L., Shah, V.H., Garcia, J.J., Tizhoosh, H.R.: Analysis and validation of image search engines in histopathology. *IEEE Reviews in Biomedical Engineering* (2024). <https://doi.org/10.1109/RBME.2024.3425769>
16. Lin, T.Y., Dollár, P., Girshick, R., et al.: Feature pyramid networks for object detection. In: *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*. pp. 2117–2125 (2017)
17. Liu, Z., Lin, Y., Cao, Y., Hu, H., Wei, Y., Zhang, Z., Lin, S., Guo, B.: Swin transformer: Hierarchical vision transformer using shifted windows. In: *Proceedings of the IEEE/CVF international conference on computer vision*. pp. 10012–10022 (2021)
18. Long, A., Yin, W., Ajanthan, T., Nguyen, V., Purkait, P., Garg, R., Blair, A., Shen, C., van den Hengel, A.: Retrieval augmented classification for long-tail visual recognition. In: *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. pp. 6959–6969 (2022)
19. Lu, M.Y., Chen, B., Williamson, D.F.K., Chen, R.J., Liang, I., Ding, T., Jaume, G., Odintsov, I., Le, L.P., Gerber, G., et al.: A visual-language foundation model for computational pathology. *Nature Medicine* **30**, 863–874 (2024). <https://doi.org/10.1038/s41591-024-02856-4>
20. Mahdizadeh Sani, M., Royat, A., Soleymani Baghshah, M.: Classification of breast cancer histopathology images using a modified supervised contrastive learning method. *Medical & Biological Engineering & Computing* (2024). <https://doi.org/10.1007/s11517-024-03225-6>
21. Naeem, A., Li, T., Liao, H.R., Xu, J., Mathew, A.M., Zhu, Z., Tan, Z., Jaiswal, A.K., Salibian, R.A., Hu, Z., Chen, T., Ding, Y.: Path-RAG: Knowledge-guided key region retrieval for open-ended pathology visual question answering. In: *Proceedings of the 4th Machine Learning for Health Symposium. Proceedings of Machine Learning Research*, vol. 259, pp. 735–746 (2025)
22. Neidlinger, P., El Nahhas, O.S.M., Muti, H.S., Lenz, T., Hoffmeister, M., Brenner, H., van Treeck, M., Langer, R., Dislich, B., Behrens, H.M., Röcken, C., Foersch, S., Truhn, D., Marra, A., Saldanha, O.L., Kather, J.N.: Benchmarking foundation models as feature extractors for weakly-supervised computational pathology. *arXiv preprint arXiv:2408.15823* (2024), <https://arxiv.org/abs/2408.15823>
23. Priyanka, V., Narendra, M., Dilleswar Rao, T.: A patient-aware benchmarking of CNN and transformer architectures for breast cancer histopathology classification. *Frontiers in Digital Health* **8**, 1752938 (2026). <https://doi.org/10.3389/fdgth.2026.1752938>
24. Spanhol, F.A., Oliveira, L.S., Petitjean, C., Heutte, L.: A dataset for breast cancer histopathological image classification. *IEEE Transactions on Biomedical Engineering* **63**(7), 1455–1462 (2016). <https://doi.org/10.1109/TBME.2015.2496264>
25. Spanhol, F.A., Oliveira, L.S., Petitjean, C., Heutte, L.: Breast cancer histopathological image classification using convolutional neural networks. In: *Proceedings of the International Joint Conference on Neural Networks (IJCNN)*. pp. 2560–2567 (2016). <https://doi.org/10.1109/IJCNN.2016.7727519>
26. Taufik, E.A., Arafath, M.A., Ghosh, A.K., Reza, M.T., Alam, M.A.: Histo-MExNet: A unified framework for real-world, cross-magnification, and trustworthy breast cancer histopathology. *arXiv preprint arXiv:2603.14416* (2026), <https://arxiv.org/abs/2603.14416>

27. Vorontsov, E., Bozkurt, A., Casson, A., Shaikovski, G., Zelechowski, M., Severson, K., Zimmermann, E., Hall, J., Tenenholtz, N., Fusi, N., Yang, E., Mathieu, P., van Eck, A., Lee, D., Viret, J., Robert, E., Wang, Y.K., Kunz, J.D., Lee, M.C.H., Fuchs, T.J.: A foundation model for clinical-grade computational pathology and rare cancers detection. *Nature Medicine* **30**(10), 2924–2935 (2024). <https://doi.org/10.1038/s41591-024-03141-0>
28. Wang, X., Du, Y., Yang, S., Zhang, J., Wang, M., Zhang, J., Yang, W., Huang, J., Han, X.: RetCCL: Clustering-guided contrastive learning for whole-slide image retrieval. *Medical Image Analysis* **83**, 102645 (2023). <https://doi.org/10.1016/j.media.2022.102645>
29. Wang, X., Yang, S., Zhang, J., Wang, M., Zhang, J.: Transformer-based unsupervised contrastive learning for histopathological image classification. *Medical Image Analysis* **81**, 102559 (2022)
30. Yu, D., Wang, Y., Jin, S., Zhu, T., Zeng, F., Qian, W., Huang, Z., Ouyang, J., Li, J., Song, Z., Guan, T., He, Y.: YpathRAG: A retrieval-augmented generation framework and benchmark for pathology. *arXiv preprint arXiv:2510.08603* (2025), <https://arxiv.org/abs/2510.08603>
31. Zhou, X., Li, C., Rahaman, M.M., Yao, Y., Ai, S., Sun, C., Li, X., Wang, Q., Jiang, T.: A comprehensive review for breast histopathology image analysis using classical and deep neural networks. *IEEE Access* **8**, 90931–90956 (2020). <https://doi.org/10.1109/ACCESS.2020.2993788>